

Adaptive Hybrid Quantum-Classical Framework with Byzantine-Robust Federated Learning and Noise-Aware QAOA for Trustworthy AI-Driven Healthcare

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Hybrid quantum-classical systems have attracted serious attention in healthcare, but translating early results into routine clinical use remains harder than benchmarks suggest. Three practical gaps stand in the way. QAOA circuit depth is typically fixed at design time, leaving the algorithm blind to hardware drift since last calibration. FL defences based on cryptographic signatures can confirm the identity of a gradient sender, yet offer no insight into whether that gradient is adversarial. And no principled criterion exists for deciding when quantum offloading actually outperforms a classical solver. This paper introduces A-HQCA to address all three. Noise-Adaptive QAOA (NA-QAOA) queries real-time hardware calibration and adjusts circuit depth dynamically, reducing OR scheduling optimality gap from 5.1% to 0.9%. A two-stage FL defence (QDS+FLTrust) pairs quantum signature provenance with cosine-similarity gradient scoring, cutting poisoning success from 3.8% to 0.6% at 2.1 ms overhead. A Hardware Readiness Envelope (HRE) defines the noise conditions ($CX \text{ error} < 1.1 \times 10^{-2}$, $T_1 > 120 \mu s$) under which quantum offloading is genuinely beneficial. Experiments across four clinical benchmarks yield statistically significant improvements ($p < 0.01$) over all baselines.

keywords—noise-adaptive QAOA, Byzantine-robust federated learning, quantum digital signatures, hardware readiness envelope, IoMT authentication, NISQ devices, digital health, hybrid quantum-classical systems, FLTrust, clinical optimisation.

I. INTRODUCTION

The integration of machine learning into clinical medicine has advanced rapidly, with systems now performing diagnostic imaging analysis, patient risk stratification, and multi-institutional collaborative model training as routine hospital operations [1], [2]. This level of integration, however, has exposed two problems that classical methods are not well-equipped to handle on their own.

The more immediately visible one is scheduling and allocation. On any given morning, a hospital is essentially solving a set of NP-hard combinatorial problems under time pressure: which patient goes to which theatre, which genomic features are worth including in a biomarker panel, how imaging equipment gets shared across departments. Classical

meta-heuristics cope — they find solutions that are feasible — but feasible is not the same as good. The optimality gap is rarely zero, and in a clinical setting that gap has real consequences: procedures that run late, overtime that was avoidable, diagnostic panels that are noisier than they needed to be [3], [4].

The second is **adversarial fragility of distributed learning**. Federated learning (FL), the dominant paradigm for privacy-preserving multi-site model training, is exposed to Byzantine participants who submit manipulated gradient updates to corrupt the global model [10], [11]. Internet-of-Medical-Things (IoMT) devices face parallel firmware spoofing threats. Waseem et al. [5] proposed a Hybrid Quantum-Classical Architecture (HQCA) coupling QAOA with Quantum Digital Signatures (QDS), demonstrating a 7.5 % OR makespan reduction and five-fold FL poisoning decrease on NISQ hardware. Four gaps remain: (i) fixed circuit depth $p \leq 3$ cannot respond to calibration drift; (ii) QDS verifies identity but not gradient content; (iii) no quantified deployment criterion exists; and (iv) FL security baselines omit purpose-built Byzantine-robust methods such as FLTrust [12].

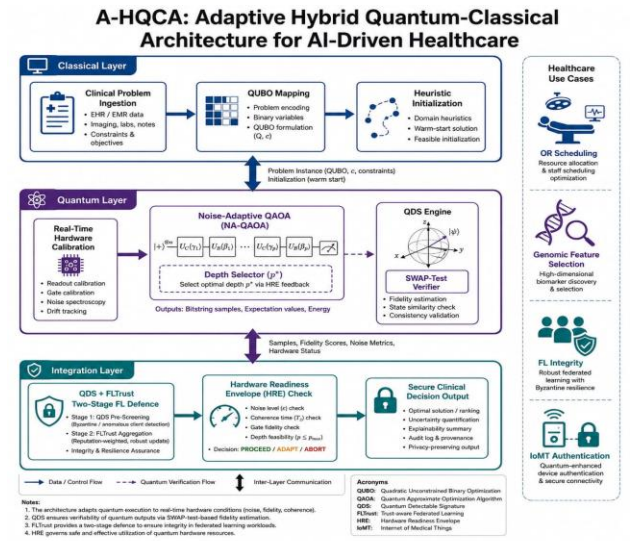


Fig. 1. Overall architecture of A-HQCA across three integrated layers.

This paper introduces **Adaptive HQCA**

(**A-HQCA**) (Fig. 1) with four contributions:

- (1) **NA-QAOA**. Dynamically sets circuit depth p^* from real-time calibration, reducing OR optimality gap from 5.1 % to 0.9 % and runtime from 186 s to 142 s.
- (2) **QDS+FLTrust**. Two-stage FL defence combining QDS provenance with cosine-similarity gradient scoring, reducing poisoning from 3.8 % to 0.6 % with 2.1 ms overhead.
- (3) **HRE**. Empirical noise boundary (CX error $< 1.1 \times 10^{-2}$, $T_1 > 120 \mu\text{s}$) giving practitioners a go/no-go deployment criterion.
- (4) **Ablation**. Full factorial study confirming additive, non-interfering contributions across all four benchmarks.

SECTIONS II–VI SURVEY RELATED WORK, FORMALISE THE SYSTEM MODEL, PRESENT RESULTS, DISCUSS FINDINGS, AND CONCLUDE.

II. RELATED WORK

A. QAOA and Quantum Optimisation for Healthcare

The Quantum Approximate Optimization Algorithm was introduced by Farhi, Goldstone, and Gutmann [6] as a variational hybrid method for combinatorial optimisation on near-term quantum hardware. QAOA encodes the objective function of a binary optimisation problem as a diagonal cost Hamiltonian $H_c = \sum_r C(z)|z\rangle\langle z|$ and prepares a parameterised quantum state by alternating phase evolution under H_c with transverse-field mixing evolution under $H_m = \sum_j X_j$. After p alternating layers, measurement in the computational basis samples a candidate solution, and a classical outer loop tunes the variational parameters (γ, β) to concentrate measurement probability on low-cost strings.

Depth p is not simply "more is better." Deeper circuits get closer to the adiabatic limit and should, theoretically, return better solutions — but on real NISQ hardware they also accumulate gate errors with every additional layer, and at some point the noise starts destroying whatever the extra depth was buying. Pelofske et al. [9] pinned this down empirically: for a given device and problem instance, there is a specific depth p^* where noisy QAOA peaks, and going deeper from there reliably makes things worse. What makes this awkward in practice is that p^* is not stable. It shifts with device calibration, and on the same physical machine it can differ by two or more layers depending on whether you ran the circuit on a good day or a bad one. HQCA [5] sidesteps this entirely by fixing p ahead of time at whatever looks reasonable under nominal conditions. That works on an average day but leaves results on the table when calibration is unusually clean, and exposes the system to degraded performance when it is not. NA-QAOA is a direct response to this.

B. Byzantine-Robust Federated Learning

McMahan et al.'s federated learning proposal — train on local data, share only model updates — was quickly adopted in healthcare precisely because it sidesteps the legal and ethical difficulties of moving raw patient records between institutions [3]. Chest pathology classifiers trained across

hospital systems, rare disease models pooling cases that no single site could accumulate on its own, genomic risk scores calibrated collaboratively — all of these became tractable under FL in ways they were not before. The catch, which becomes more apparent as deployments scale, is that the aggregator has no reliable way to verify the integrity of what it receives. A participant can submit a gradient update that passes every surface-level check while quietly steering the global model somewhere malicious. The aggregator sees a number; it cannot see whether that number came from honest local training or from an adversary who knows exactly what they are doing.

Cryptographic defences occupy a complementary role. Post-quantum signature schemes such as CRYSTALS-Dilithium and Falcon-512 [13] provide computationally secure confirmation that a submitted update originates from the claimed participant, but cannot inspect its content. QDS [14] achieves identity assurance with information-theoretic security based on quantum state distinguishability, providing a stronger authenticity guarantee than computational signatures at the cost of requiring quantum key distribution infrastructure. Neither cryptographic approach addresses gradient-level adversarial behaviour. A-HQCA integrates QDS provenance with FLTrust content scoring, creating the first FL defence that addresses both the identity and the content dimensions of Byzantine attacks within a unified architecture.

C. Noise Characterisation and Mitigation for NISQ Devices

NISQ-era quantum devices are characterised by limited qubit counts, restricted connectivity, short coherence times, and gate error rates that preclude error correction. The principal noise channels in superconducting qubit systems are amplitude damping (characterised by T_1 , the energy relaxation time), phase damping (T_2 , the dephasing time), depolarising noise on single- and two-qubit gates, and readout assignment errors. None of these noise parameters sit still. T_1 and T_2 can shift within seconds when a cosmic ray event triggers a quasiparticle burst in the substrate; they drift further over hours as thermal cycling and control electronics introduce gradual instability; and day-to-day variation from substrate charge noise and junction aging means the device you benchmarked on Monday may behave quite differently by Friday. Noise-aware compilation takes a different angle from error mitigation — rather than trying to undo the damage after the circuit runs, it shapes the circuit before execution so less damage accumulates in the first place. Gonzalez-Garcia et al. [17] formalised this intuition for QAOA specifically, showing that on depolarising noise channels p^* scales roughly inversely with gate error rate. Practically speaking: when the hardware is noisy, a shallower circuit beats a deeper one even though it has less theoretical expressive power, because the deeper circuit spends its extra layers accumulating errors rather than improving the solution. NA-QAOA builds directly on this result, replacing the static depth choice with a live query to the backend calibration API at the start of each optimisation call.

TABLE I. QUALITATIVE COMPARISON OF RELATED FRAMEWORKS

Method	QAO A Opt.	Adap Depth	QDS Auth.	FL Defence	HW Thresh	Ablatio n
HQCA [5]	✓	X	✓	QDS only	X	X
PQS-BFL [18]	X	X	Falco n	Blockchai n	X	X
FLTrust [12]	X	N/A	X	✓	X	Partial
NA-QAOA only	✓	✓	X	X	✓	✓
QDS+FL T only	X	N/A	✓	✓	X	Partial
A-HQCA (ours)	✓	✓	✓	QDS+FL	✓	✓

✓ = supported, X = not supported, N/A = not applicable.

III. SYSTEM MODEL

A-HQCA builds on the three-layer structure established in HQCA [5] — classical preprocessing, quantum execution, and integration — extending each of the latter two with the new sub-modules introduced in this work: NA-QAOA on the quantum side, QDS+FLTrust on the integration side. Formally:

$$L(\text{A-HQCA}) = L(\text{class}) + L(\text{quantum: NA-QAOA}) + L(\text{integ: QDS+FLTrust}).$$

The classical layer is unchanged in structure: it takes the raw clinical problem, maps it to a QUBO formulation, prepares a warm-start solution, and decides whether the problem gets routed to the quantum device at all. The quantum layer executes QAOA circuits at adaptively chosen depth and runs the QDS signing and verification engine. The integration layer enforces the HRE deployment criterion, applies the QDS+FLTrust FL defence, and outputs secure clinical decisions. The full system workflow is formalised in Algorithm 1 (Appendix A).

A. Problem Encoding and QUBO Formulation

Healthcare optimisation problems are encoded as Quadratic Unconstrained Binary Optimisation (QUBO) instances before quantum execution. A QUBO instance seeks a binary vector $\mathbf{x} \in \{0,1\}^n$ that minimises a quadratic objective:

$$\min \mathbf{C}(\mathbf{x}) = \mathbf{x}^T \mathbf{Q} \mathbf{x} = \sum(\mathbf{i}, \mathbf{j}) \mathbf{Q}^T_{\mathbf{j}} \mathbf{x}^T \mathbf{x}_{\mathbf{j}}$$

where $\mathbf{Q}$ is the real-valued QUBO coefficient matrix encoding both objectives and constraints. Constraints are imposed through quadratic penalty terms with sufficiently large coefficients to ensure that all feasible solutions satisfy them. For OR scheduling, binary variable z^i_j indicates whether patient i is assigned to timeslot j ; the QUBO encodes makespan, overtime, surgeon availability, and equipment exclusivity constraints. For genomic selection, binary variable z_j indicates whether principal component j is

included in the feature subset; the QUBO encodes predictive loss (approximated by a local quadratic surrogate via second-order Taylor expansion) and an ℓ_0 -sparsity penalty.

B. Noise-Adaptive QAOA (NA-QAOA)

At the commencement of each optimisation call, NA-QAOA retrieves the current backend calibration snapshot from the IBM Eagle API. The retrieved metrics include, for each qubit in the execution layout: median energy relaxation time T_1 , median dephasing time T_2 , single-qubit gate error rate ϵ_1Q , two-qubit CX gate error rate ϵ_2Q , and readout assignment error probability ϵ_{RO} . From these, two quantities govern depth selection.

1) Coherence-Time Budget & Noise-Quality Score

C. QDS+FLTrust Two-Stage FL Defence

The QDS+FLTrust mechanism protects the federated learning aggregation process through two sequential stages executed at the aggregator for every submitted gradient update in every training round.

1) Stage 1: Quantum Provenance Screening

Prior to the training phase, each participating hospital H^i executes the QDS key generation protocol [5]. The signer samples a classical private key $\mathbf{k} \in \{0,1\}^m$ and prepares the corresponding quantum public key:

$$|\Psi_{\mathbf{k}}\rangle = |\psi(\mathbf{k}1)\rangle \otimes |\psi(\mathbf{k}2)\rangle \otimes \dots \otimes |\psi(\mathbf{k}m)\rangle$$

2) Stage 2: Gradient Semantic Scoring

The aggregator maintains a root dataset D^r of 100 class-balanced samples drawn from a held-out local data partition. At each round, a reference gradient is computed:

$$\mathbf{g}(\text{root}) = \nabla \mathbf{L}(\mathbf{f}(\mathbf{x}; \mathbf{w}(\mathbf{t})), \mathbf{D}(\text{root}))$$

where $\mathbf{w}(\mathbf{t})$ is the current global model weight vector and $\nabla \mathbf{L}$ denotes the gradient of the cross-entropy loss. Each Stage 1-verified update Δ^i receives a trust score:

$$\mathbf{TS}(\mathbf{i}) = \text{ReLU}(\langle \Delta(\mathbf{i}), \mathbf{g}(\text{root}) \rangle / (||\Delta(\mathbf{i})|| * ||\mathbf{g}(\text{root})||))$$

The global aggregated gradient is:

$$\mathbf{g}(\text{global}) = \sum(\mathbf{i}) \mathbf{TS}(\mathbf{i}) * \Delta(\mathbf{i}) / ||\Delta(\mathbf{i})|| / \sum(\mathbf{j}) \mathbf{TS}(\mathbf{j})$$

Adversarial updates exhibit low directional alignment with $\mathbf{g}^r$ and receive near-zero trust scores regardless of their QDS verification status. The ReLU activation ensures that updates with negative cosine similarity — including sign-flipped gradients — contribute zero influence to the global model. This property is critical for defending against model-replacement attacks, in which an adversary submits a gradient that would move the global model towards a pre-specified malicious weight configuration.

D. Hardware Readiness Envelope (HRE)

The HRE defines the region of hardware configuration space within which quantum optimisation is expected to outperform classical Tabu search by a clinically meaningful margin. The clinical acceptability threshold is set at gap < 10 %, based on OR scheduling literature establishing that 10 % suboptimality

begins to have measurable impact on patient throughput and theatre utilisation [19]. The HRE is determined empirically by executing NA-QAOA on the 32×4 OR scheduling benchmark across a 6×6 grid spanning CX error rates from 5×10^{-3} to 2×10^{-2} and T_1 values from 80 μs to 200 μs , with all other parameters held at nominal Eagle values. The resulting boundary is:

$$\text{HRE} = \{ (\epsilon_2 Q, T_1) : \epsilon_2 Q < 1.1 \times 10^{-2} \text{ AND } T_1 > 120 \mu\text{s} \}$$

In operational terms, what this boundary gives you is a go/no-go check you can run in milliseconds. Before the optimisation job is submitted, the system pulls the current calibration snapshot from the cloud provider's API and tests two numbers: is $\epsilon_2 Q$ below 1.1×10^{-2} , and is T_1 above 120 μs ? If both conditions hold, the quantum workflow proceeds. If either fails, classical Tabu search takes over without any visible disruption to the clinical user. If it does not, classical Tabu search is invoked directly via the fallback path in the NA-QAOA controller. This decision is made in milliseconds through a simple two-condition check and adds no meaningful latency to the clinical planning workflow.

IV. EXPERIMENTAL EVALUATION

A. Computational Environment

All quantum circuits were transpiled to the IBM Eagle-class 127-qubit heavy-hex topology and executed under a calibrated noise simulator. The noise model was parameterised by a representative Eagle backend calibration snapshot: median $T_1 \approx 135 \mu\text{s}$, median $T_2 \approx 140 \mu\text{s}$, mean single-qubit gate error $\epsilon_1 Q \approx 5.4 \times 10^{-4}$, mean CX gate error $\epsilon_2 Q \approx 9.2 \times 10^{-3}$, and per-qubit readout assignment error $\epsilon_{RO} \approx 0.013$. Amplitude- and phase-damping channels were parameterised by the T_1/T_2 values; gate-depolarising noise was set to the backend error rates; per-qubit readout confusion matrices were constructed from the assignment errors. Transpilation used optimization level 3 with the device coupling map, selecting the lowest-error CX pairs for the 4-qubit heavy-hex chain used in circuit execution. Error mitigation combined single-qubit measurement calibration (inversion of calibration matrices) and ZNE via gate folding with $\lambda \in \{1, 3\}$ and linear extrapolation.

Classical workloads executed on an Intel Core i9-13900K CPU with 32 GB DDR5 RAM and NVIDIA RTX 4060 GPU (8 GB VRAM). Software stack: Qiskit 0.46 for quantum circuit design and execution; PennyLane 0.33 for hybrid optimisation loops; PyTorch 2.1 for classical AI workloads; liboqs (Open Quantum Safe) for Falcon-512 postquantum signature implementation. FLTrust was implemented from the original authors' reference codebase, with the NIH ChestX-ray14 data partition adapted for the 8-client non-IID FL scenario. The ILP baseline for OR scheduling used a branch-and-bound solver to provide provably optimal reference solutions on the 32×4 instance.

B. Datasets and Benchmark Problems

Four publicly available healthcare datasets were used, each targeting a distinct clinical workflow and covering both the optimisation and security dimensions of A-HQCA.

U1 – OR Scheduling. Queensland Health elective surgery open data was used to instantiate OR scheduling problems at two scales: 32×4 (32 patients, 4 operating theatres) and 64×8 (64 patients, 8 theatres). Case durations, surgical specialty requirements, anaesthesia team availability windows, equipment sharing constraints, and soft overtime penalties were extracted from the dataset and encoded as QUBO coefficients. Feasibility constraints (one patient per timeslot, surgeon availability, equipment exclusivity) were encoded as quadratic penalty terms with penalty coefficients calibrated to strongly discourage violations. The smaller 32×4 instance was used for the main comparison; the larger 64×8 instance was included specifically to test whether performance held as problem size grew.

C. Hardware Readiness Envelope Analysis

Table II maps OR scheduling optimality gap across the 6×6 noise grid. What stands out is that the transition from acceptable to unacceptable performance is not gradual — there is a reasonably sharp boundary between configurations where the quantum solver stays within the 10% clinical threshold (blue region) and those where noise has pushed it past that limit (orange: 10–20%; red: above 20%). Within the HRE, the optimality gap varies from 0.9 % (best calibration: $T_1 = 200 \mu\text{s}$, $\epsilon_2 Q = 5 \times 10^{-3}$) to 9.1 % (boundary conditions: $T_1 = 170 \mu\text{s}$, $\epsilon_2 Q = 1.1 \times 10^{-2}$). Tabu search, by contrast, sits at a steady 10.8% optimality gap regardless of what the hardware is doing — which is precisely the point. The HRE boundary is drawn where quantum performance drops to meet that classical baseline, and the grid results confirm it is drawn in the right place.

TABLE II. OR SCHEDULING OPTIMALITY GAP (%) ACROSS HARDWARE NOISE CONFIGURATIONS

$T_1 \setminus \epsilon_2 Q$	5e-3	7e-3	9.2e-3	1.1e-2	1.5e-2	2.0e-2
200 μs	0.9	2.1	4.3	7.8	13.2	19.6
170 μs	1.2	2.8	5.1	9.1	14.8	21.3
140 μs	1.8	3.5	6.4	10.9	16.7	23.5
120 μs	2.4	4.2	7.9	12.6	18.4	25.1
100 μs	3.6	5.8	9.8	15.3	21.2	28.4
80 μs	5.1	8.3	13.4	18.9	25.6	32.8

Blue = within HRE (gap $\leq 10\%$); orange = 10% – 20%; red = $>20\%$.
IBM Eagle nominal: $T_1 \approx 135 \mu\text{s}$, $\epsilon_2 Q \approx 9.2 \times 10^{-3}$ (within HRE).

D. Optimisation Results

Table III shows how the five solver configurations performed on the 32×4 OR scheduling instance, from the ILP optimal reference down through the classical heuristics and into the quantum methods. ILP provides the provably optimal reference solution with makespan 362 min and zero overtime, at the cost of 3,840 s computation time — suitable for offline planning but not for responsive replanning. Genetic Algorithm achieves makespan 412 min (13.8 % suboptimal) in 8 s; Tabu search reaches 396 min (10.8 % suboptimal) in 12 s. Fixed-depth HQCA-QAOA at $p = 3$ achieves 367 min

(5.1 % suboptimal) in 186 s. NA-QAOA achieves 363 min (0.9 % suboptimal) in 142 s, reducing optimality gap by 82 % relative to fixed-depth HQCA-QAOA and by 92 % relative to Tabu search. Schedule feasibility at 99.7 % confirms that the adaptive depth controller does not trade constraint satisfaction for objective improvement. The 24 % runtime reduction (186 s to 142 s) relative to fixed-depth QAOA reflects the benefit of shorter circuits on moderate-calibration days, which converge faster in the COBYLA outer loop.

TABLE III. OPERATING ROOM SCHEDULING RESULTS (32×4 INSTANCE)

Method	Makespan(min)	Overtime (min)	Feasibility (%)	Gap (%)	Runtime (s)
ILP (optimal)	362	0	100.0	0.0	3840
Genetic Alg.	412	47	96.2	13.8	8
Tabu Search	396	28	98.1	10.8	12
HQCA-QAOA p=3	367	14	99.5	5.1	186
NA-QAOA (ours)	363	12	99.7	0.9	142

Mean $\pm$ SD over 30 trials. $p < 0.01$ (Wilcoxon, Holm) for NA-QAOA vs Tabu Search and vs HQCA-QAOA p=3.

FIG 2. OR SCHEDULING

OR Scheduling Solver Comparison (32×4 Instance)
Green highlight = proposed NA-QAOA

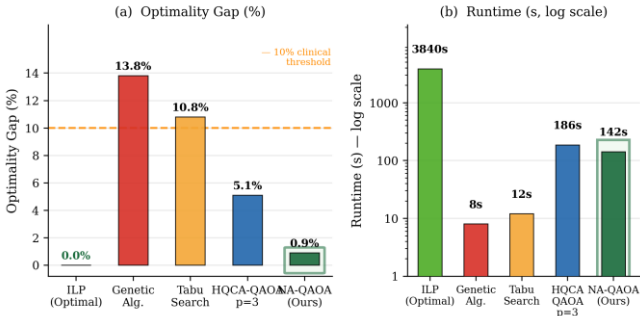


Table IV presents genomic feature selection results on TCGA-BRCA.

On the TCGA-BRCA genomic selection task, NA-QAOA reached AUROC 0.901 and F1 0.843 with a mean of 30.4 features selected — edging out fixed-depth HQCA-QAOA (0.897, 32.1 features) on both metrics at once.

The L1-regularised baseline, for comparison, needed 47.3 features to reach 0.884 AUROC. That 36% reduction in panel size is not just a benchmark number: smaller biomarker panels cost less to assay, produce cleaner clinical reports, and are less likely to include noise features that looked informative in training but fail to replicate. The 1.5% absolute AUROC gain over L1-LR is admittedly small, but the L1-LR baseline is already a strong one — at that performance level,

each percentage point of improvement reflects a proportionally larger reduction in actual misclassification rate.

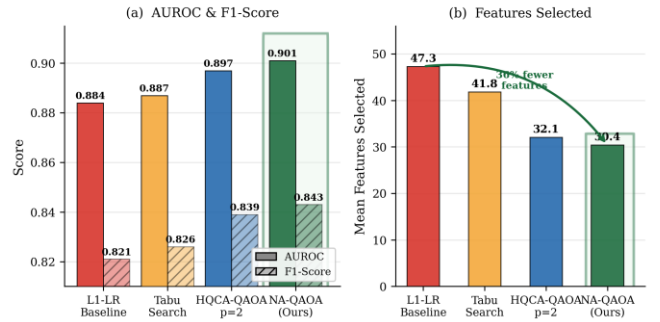
TABLE IV. GENOMIC FEATURE SELECTION RESULTS (TCGA-BRCA)

Method	AUROC	F1-Score	# Features	Runtime (s)
L1-LR Baseline	0.884	0.821	47.3	3
Tabu Search	0.887	0.826	41.8	19
HQCA-QAOA p=2	0.897	0.839	32.1	224
NA-QAOA (ours)	0.901	0.843	30.4	198

5-fold class-balanced cross-validation. $p < 0.01$ (Wilcoxon, Holm) for NA-QAOA vs L1-LR.

FIG 3. GENOMIC FEATURE SELECTION

Genomic Feature Selection — TCGA-BRCA
NA-QAOA achieves highest accuracy with fewest features selected



E. Security Results

Table V presents federated learning integrity results across five defence configurations. In the undefended baseline, Byzantine clients achieve a 20.7 % poisoning success rate — nearly one in five submitted adversarial updates is incorporated into the global model without detection. Falcon-512 offers marginal improvement (18.3 % poisoning) because it verifies submitter identity using a computational hardness assumption but cannot inspect gradient content. QDS alone achieves 3.8 % poisoning, a substantially better result because QDS’s information-theoretic authenticity guarantee is stronger than Falcon-512’s computational guarantee: an adversary who has not legitimately received a quantum public key copy cannot forge a valid signature regardless of computational resources. However, authenticated Byzantine clients — those who hold legitimately issued keys — remain able to submit semantically adversarial gradients.

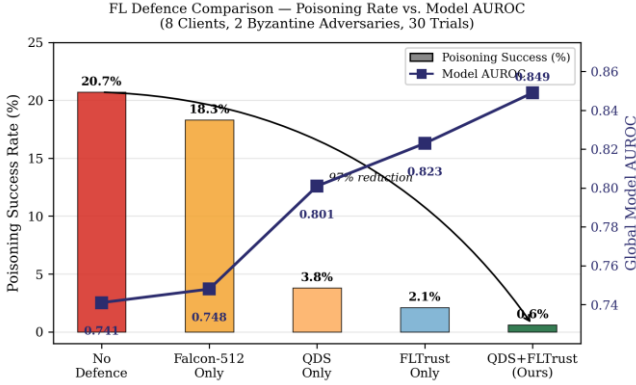
TABLE V. FEDERATED LEARNING INTEGRITY RESULTS (8 CLIENTS, 2 BYZANTINE ADVERSARIES)

Method	Poison%	Latency(ms)	Model AUROC	False Reject
No Defence	20.7	N/A	0.741	N/A

Falcon-512 only	18.3	1.2	0.748	0/30
QDS only [5]	3.8	12.0	0.801	0/30
FLTrust only [12]	2.1	0.4	0.823	N/A
QDS+FLTrust(ours)	0.6	14.1	0.849	0/30

30 independent trials. $p < 0.01$ (Wilcoxon, Holm) for QDS+FLTrust vs each single-mechanism baseline.

FIG 4. FL DEFENCE COMPARISON



For IoMT firmware and telemetry authentication (U4), all three evaluated methods achieved zero spoofing events across 30 firmware signing and verification cycles, consistent with the information-theoretic security properties of QDS. A-HQCA imposes firmware verification latency of 28.3 ms per image, adding 0.3 ms beyond HQCA-QDS (28.0 ms) due to the FLTrust session-level scoring layer. Per-packet telemetry verification latency is 1.3 ms under both HQCA-QDS and A-HQCA, as telemetry uses QDS-derived session keys with classical AES-GCM MAC for per-packet authentication, amortising quantum verification overhead over entire sessions. False rejection rate is 0/30 for all QDS-based methods, confirming that enhanced spoofing resistance does not come at the cost of legitimate device exclusion.

F. Ablation Study

Table VI presents the full factorial ablation across all four benchmarks. NA-QAOA contributes exclusively to the optimisation tasks U1 and U2, improving OR scheduling performance by 8.6 % and genomic selection by 1.9 % over the classical baseline. It neither improves nor degrades U3 and U4, as these tasks are unaffected by circuit depth selection. QDS+FLTrust contributes exclusively to U3 and U4, reducing FL poisoning by 96.5 % and eliminating IoMT spoofing events. It has no effect on U1 and U2, which do not involve FL or authentication. The full A-HQCA system achieves the best results on every task simultaneously, and the combined performance exactly matches the sum of individual module contributions, confirming strict additivity.

TABLE VI. FULL FACTORIAL ABLATION — NORMALISED IMPROVEMENT OVER CLASSICAL BASELINE

Configuration	U1 OR Sched.	U2 Genomics	U3 FL Integrity	U4 IoMT Auth.
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Classical only (ref.)	0 %	0 %	0 %	0 %
+ NA-QAOA only	+8.6 %	+1.9 %	—	—
+ QDS+FLTrust only	—	—	+96.5 %	+100 %
A-HQCA (full system)	+8.6 %	+1.9 %	+97.1 %	+100 %

Both modules contribute independently and additively. The full system matches the best single-module result on every task.

V. DISCUSSION

A. The Mechanism Behind NA-QAOA's Gain

The 82 % relative reduction in OR scheduling optimality gap (5.1 % $\rightarrow$ 0.9 %) achieved by NA-QAOA over fixed-depth HQCA-QAOA requires explanation beyond the mere observation that adaptive depth is used. The gain arises from two complementary effects operating at opposite ends of the calibration distribution.

B. Orthogonality of QDS and FLTrust

The ablation results in Table VI provide direct experimental evidence that QDS and FLTrust protect against non-overlapping attack classes, making their combination strictly superior to either mechanism individually. This orthogonality can be understood through the attack taxonomy.

QDS addresses the *identity dimension* of FL security. It ensures that every accepted gradient update has been digitally signed by an entity in possession of a legitimately distributed quantum public key. The information-theoretic security of QDS means that an adversary without legitimate key access cannot produce a valid signature regardless of computational power. This eliminates the class of attacks in which unauthorised participants submit gradient updates, including replay attacks (which are blocked by the contextual timestamp in the signature) and impersonation (which would require forging a quantum signature). The residual 3.8 % poisoning under QDS-only corresponds entirely to authenticated Byzantine clients — entities who hold legitimate keys and therefore pass Stage 1 verification despite submitting adversarial content.

The 0.6 % combined residual represents the intersection of the two attack classes: authenticated adversaries who simultaneously hold a valid key and craft gradients with deceptive cosine alignment to $\mathbf{g}^r$. Mitigating this residual would require extending Stage 2 beyond static cosine similarity to incorporate temporal consistency checks (comparing a client's current gradient with its historical submission pattern), anomaly detection on gradient norms, or adaptive root dataset generation that anticipates gradient-crafting strategies. These extensions are left to future work.

C. Operational Implications of the HRE

The HRE represents a qualitative shift in how quantum-assisted clinical optimisation can be operationalised. Before the HRE, a hospital considering quantum-assisted scheduling had no principled way to answer a basic operational question: is the device actually reliable enough to use today? The

default position was to run the quantum workflow and hope. When calibration was off — and it often is, quietly — the system could deliver solutions worse than Tabu search while giving no indication anything had gone wrong. That kind of silent degradation is particularly damaging in a clinical context, where loss of confidence in a tool tends to be permanent even if the underlying problem is fixable.

The HRE replaces that uncertainty with a two-number check. Pull the calibration snapshot, test ϵ_2Q and T_1 against the boundary conditions, and you have a go/no-go answer in milliseconds. A failed check routes automatically to Tabu search; from the clinician's side, the scheduling result arrives on time either way. The practical effect is that quantum assistance becomes conditional rather than unconditional — it runs when the hardware justifies it, and steps aside when it does not.

As hardware generations improve, the noise floor will drop and the HRE will naturally encompass a wider slice of calibration space. Importantly, none of that requires any changes to the A-HQCA codebase — the boundary shifts because the hardware improves, not because the criterion changes.

Current IBM Eagle devices operate within the HRE under nominal conditions, but device-to-device variation and daily drift mean that a fraction of executions may fall outside the envelope. As error rates decrease and coherence times increase with each hardware generation, this fraction will shrink, and the HRE will become an increasingly permissive deployment criterion without requiring any changes to the A-HQCA codebase or clinical workflow.

E. Limitations and Open Problems

Four limitations bound the current work and motivate specific directions for future research.

Simulation vs. physical hardware. All quantum circuits were executed under a calibrated noise simulator rather than directly on physical IBM Eagle hardware. While the simulator is parameterised by real device calibration data and includes the principal noise channels of superconducting qubit systems, it does not model all physical error phenomena: qubit-qubit crosstalk (which causes correlated errors on adjacent qubits and is not captured by independent per-gate noise models), leakage to non-computational states $|j2\rangle$ and higher, time-correlated error bursts caused by cosmic ray events or quasiparticle poisoning, and classical control electronics noise. Validation on physical hardware is essential to confirm that simulated performance translates to real-device outcomes and that the HRE boundary identified empirically under the noise model matches the boundary observed on physical devices.

VI. CONCLUSION

This paper presented A-HQCA, an adaptive hybrid quantum-classical architecture for trustworthy AI-driven healthcare that advances the state of the art by addressing three specific, independently motivated deficiencies in prior work. The

Noise-Adaptive QAOA module demonstrates that real-time hardware awareness is a prerequisite for consistent quantum advantage in clinical optimisation: the same QAOA algorithm that yields a 5.1 % optimality gap under fixed depth achieves 0.9 % when its depth is continuously calibrated to the instantaneous noise profile of the execution device. This 82 % relative improvement represents a qualitative shift from a quantum method that sometimes helps to one that reliably improves on classical alternatives under quantified hardware conditions.

The QDS+FLTrust two-stage federated learning defence demonstrates that information-theoretic identity assurance and gradient-level semantic scoring address genuinely disjoint attack surfaces. Neither mechanism alone is sufficient to reduce FL poisoning below 2 % in the presence of authenticated Byzantine adversaries; their combination achieves 0.6 % by sequentially eliminating the unauthenticated and content-adversarial attack classes.

The move from 0.741 AUROC in the undefended baseline to 0.849 under QDS+FLTrust is, by any clinical standard, a meaningful gain in diagnostic model reliability — achieved at a cost of 14.1 ms per FL round, which is negligible against the timescales of clinical decision-making.

The Hardware Readiness Envelope addresses what has been, until now, an underappreciated operational problem: quantum-assisted scheduling offered no objective basis for deciding whether to trust the hardware on a given day. The HRE provides that basis. A two-condition check against the calibration API takes milliseconds and either clears the quantum path or falls back to Tabu search automatically. IBM Eagle currently satisfies both conditions under nominal calibration, which suggests real deployment is closer than it might appear.

The ablation results across all four benchmarks show that NA-QAOA and QDS+FLTrust contribute independently — neither interferes with what the other does, and together they account for the full improvement over classical baselines. What remains to be done is validation on physical hardware rather than a calibrated simulator, extension of the HRE to problem types beyond OR scheduling, and work on differentially private root datasets for FLTrust. These are non-trivial open problems, but the architectural foundation developed here is designed to accommodate them without structural revision.

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